A Computable Map of Treatment-Sequencing Evidence in HR+/HER2- Metastatic Breast Cancer: A Graph-Theoretic Pilot Comparing Trial Chains with the ESMO Guideline Decision Tree

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Abstract

Purpose. HR+/HER2- metastatic breast cancer (mBC) guidelines compose multi-line treatment recommendations across non-overlapping pivotal trials, but the structural concordance between guideline decision trees and the underlying trial evidence has not been quantified. We introduce a graph-theoretic framework that makes this concordance computable and apply it to the ESMO 2024 update. **Methods.** We retrieved 16 trials in total — 15 pivotal phase 2/3 HR+/HER2- mBC trials and one PARP-inhibitor reference trial for the gBRCAm decision node — from ClinicalTrials.gov v2 (freeze 2026-05-16) and structured each trial's eligibility into a frozen JSON schema by hand-curated extraction from canonical primary publications. The corpus was represented as a directed graph whose (prior-state, biomarker) nodes are shared with a parallel 16-node encoding of the ESMO HR+/HER2- decision tree. The evidence-free decision-point ratio (EFDPR) is the fraction of guideline nodes with no trial edge satisfying state-superset, biomarker-concordance, drug-class, and temporal-precedence criteria, reported under three pre-registered tolerance levels (strict, ESCAT-aligned, liberal). The pre-registered primary test was a one-sided exact binomial test of H_0 : EFDPR ≤ 0.25; Clopper-Pearson and bootstrap-percentile 95% CIs accompany each estimate. **Results.** Under strict and ESCAT-aligned concordance the EFDPR was 0.31 (5/16; Clopper-Pearson 95% CI 0.11–0.59); liberal tolerance gave 0.19 (3/16; CI 0.04–0.46). The pre-registered exact-binomial test failed to reject the null under all three tolerance levels (strict and ESCAT $P = 0.37$; liberal $P = 0.80$). Evidence-free nodes under strict concordance clustered at post-CDK4/6i and adjacent biomarker-stratified positions (ESR1mut, AKT-pathway, post-CDK4/6i PIK3CAmut, post-CDK4/6i no-actionable-mutation, and the post-CDK4/6i+post-endo everolimus node). Operationalization discordance was highest for the prior-CDK4/6i inclusion variable (ODI 0.91, bootstrap 95% CI 0.76–1.00; 7 trials). **Conclusion.** In this pre-registered 16-trial pilot the EFDPR point estimate modestly exceeded 0.25 but the pre-registered exact-binomial test did not reject at $\alpha = 0.05$; the result is best read as descriptive evidence of composition-across-non-overlapping-trials in post-CDK4/6i biomarker-stratified positions, with confirmatory testing deferred to a systematic-review-scale extension. The framework, schema, decision-tree encoding, and code are released under MIT licence.

Key Objective. To make computable, with a reproducible pipeline, where ESMO HR+/HER2- metastatic breast cancer guideline recommendations lack directly supporting trial evidence chains.

Knowledge Generated. A graph-theoretic framework that places trial corpora and guideline trees on the same (state, biomarker) node space and computes an evidence-free decision-point ratio (EFDPR). In a 16-trial pilot, strict EFDPR was 0.31 (Clopper-Pearson 95% CI 0.11–0.59); the pre-registered exact-binomial test did not reject ($P = 0.37$). Evidence-free nodes clustered at post-CDK4/6i biomarker-stratified positions.

Relevance. Guideline committees and trial sponsors can adopt the EFDPR overlay as a guideline-update audit, and target the next pivotal-trial agenda toward the decision nodes that current evidence chains do not cover.

Keywords: metastatic breast cancer; clinical guidelines; evidence synthesis; directed acyclic graph; computational meta-research; CDK4/6 inhibitor.

1 Introduction

Over the past decade the HR+/HER2-negative metastatic breast cancer (mBC) treatment landscape has expanded from a small number of endocrine and chemotherapeutic options to more than fifteen approved systemic regimens, including CDK4/6 inhibitors, selective estrogen receptor degraders, PI3K/AKT-pathway inhibitors, and HER2-low targeted antibody-drug conjugates.[1–3] Most of these agents were approved on the basis of pivotal trials that fix a single point in the multi-line treatment trajectory — typically post-endocrine and post-CDK4/6i second- or third-line — without head-to-head testing against the other agents that are simultaneously approved for closely overlapping populations.

Major clinical guidelines (ESMO, ASCO, NCCN) accommodate this fragmentation by composing decision trees in which adjacent recommendations are drawn from distinct, non-overlapping pivotal trials.[1–3] The legitimacy of these compositions has been argued narratively[4,5] but has not been measured as a structural property of the trial corpus. As a consequence, neither guideline committees nor trial sponsors have a quantitative answer to a basic question: which guideline-recommended decision nodes are supported by a trial chain that actually traverses them, and which are supported only by composition across non-overlapping trials?

We address this gap by introducing a graph-theoretic representation in which the pivotal-trial corpus and the guideline decision tree share a common node space defined by (prior-treatment-state, biomarker-profile) tuples. Each pivotal trial becomes a directed edge between nodes; each guideline recommendation becomes a labelled node-to-treatment mapping. With both objects on the same graph, the proportion of guideline decision nodes that admit no concordant trial edge — the *evidence-free decision-point ratio* (EFDPR) — becomes directly computable. We apply this framework to HR+/HER2- mBC under the ESMO 2024 update, pre-register the primary outcome and its rejection threshold, and release the full pipeline, schema, and decision-tree encoding for reuse.

Our contributions are: (i) a frozen JSON schema for trial-eligibility structuring across mBC, (ii) a concordance algorithm that jointly enforces patient-state, biomarker-definition, and temporal-precedence criteria, (iii) a pre-registered EFDPR estimate with bootstrap confidence interval for ESMO HR+/HER2- mBC, and (iv) a quantification of operationalization discordance for the biomarker variables most consequential to current guideline recommendations.

2 Methods

2.1 Data sources and freeze dates

The pivotal-trial corpus was retrieved from ClinicalTrials.gov via API v2 on 2026-05-16, restricted to phase 2/3 interventional trials with at least one breast-cancer condition tag, HR+ or estrogen-receptor-positive enrolment criteria, HER2-negative or HER2-low enrolment criteria, and a metastatic or advanced-disease indication. The ESMO mBC 2024 update was retrieved through its Annals of Oncology DOI version; the ASCO mBC living guideline was retrieved from open access; NCCN was used only in sensitivity analysis.

2.2 Trial-eligibility structuring

A frozen JSON schema (v1.0; Supplementary Table S2) captures, for each trial, the required and excluded prior treatments (endocrine, CDK4/6i, chemotherapy lines in the metastatic setting) and the biomarker requirements (HR status, HER2 status with operational definition source, PIK3CA / ESR1 / AKT-pathway / BRCA / HRD status), and where relevant the trial's registrational biomarker subgroup readouts. For this pilot, eligibility coding was performed by hand-curated assignment from each trial's canonical primary publication and verified against the ClinicalTrials.gov eligibility text; the per-trial mapping is published as Supplementary Table S2. A production-grade LLM-extraction pipeline with chain-of-thought prompts and inter-rater agreement against this hand-curated set is the subject of a companion methods paper and is not used to generate the present results.

2.3 DAG construction

Nodes were defined as the canonical hashed concatenation of binary prior-state flags (`post-endo`, `post-CDK46i`, `post-chemo`) and binary biomarker flags (`HR+/HER2-`, `HR+/HER2low`, `PIK3CAmut`, `ESR1mut`, `AKTpath`). For each trial, the source node encodes the required prior state at enrolment and the target node encodes the post-trial state (source state $\cup$ {drug class tested}). Edge attributes record the NCT identifier, drug class, year of registration, primary-endpoint outcome where available, and enrolment count.

2.4 Guideline decision-tree encoding

ESMO and ASCO decision points were structured by manual annotation plus LLM-assisted draft, then reviewed against the source guideline text. Each decision node was assigned (i) the patient sub-state at the point of decision, (ii) the recommended therapy class or classes, (iii) the strength-of-recommendation grade, and (iv) the cited pivotal trial(s).

2.5 Concordance algorithm

For each guideline node g, the supporting-evidence set Supp(g) is the set of trial-DAG edges (u, v, t) that simultaneously satisfy: (a) *state superset* — every guideline-required prior-treatment token of g is a member of the trial's source-state token set, with the special line-agnostic token `metastatic` (used for gBRCAm) satisfied by any state (this token is a disclosed post-hoc addition to the pre-registered schema; see Supplementary Text 1); (b) *drug match* — the trial drug class is identical to, or a member of an explicit equivalence group with, one of the guideline-recommended classes at g (the equivalence groups are listed in the source code at `analysis/05_compute_efdpr.py`); (c) *biomarker concordance* under one of three pre-registered tolerance levels (strict: exact set match; ESCAT-aligned: strict plus ESCAT-equivalence groupings; liberal: ESCAT plus acceptance of registrational biomarker-subgroup readouts); (d) *temporal precedence* — the trial primary-completion year does not post-date the guideline-version year. The evidence-free decision-point ratio is

$$\mathrm{EFDPR} = \frac{|\{g : \mathrm{Supp}(g) = \emptyset\}|}{|\mathcal{G}|},$$

where $\mathcal{G}$ is the set of ESMO HR+/HER2- mBC decision nodes used for inference.

2.6 Operationalization discordance index (ODI)

For each biomarker variable b shared by multiple trials, the ODI is the mean pairwise Jaccard distance between the trial-level inclusion definitions of b, treating each definition as a set of operational tokens. The token vocabulary is hand-curated against each trial's canonical primary publication and is published as Supplementary Table S4; a future LLM-extraction-based replacement will produce the same vocabulary automatically and will be validated against this hand-curated set.

2.7 Statistical inference and pre-registration

The primary outcome (EFDPR), primary hypothesis (EFDPR > 0.25), rejection threshold ($\alpha = 0.05$, one-sided exact binomial), secondary outcomes, and stop rules were committed to `docs/prereg.md` before any outcome-touching analysis was run. The pre-registered primary test is the one-sided exact binomial test of H_0 : EFDPR ≤ 0.25 given k evidence-free nodes out of $n = |\mathcal{G}|$ guideline nodes. Two-sided 95% confidence intervals are reported in two forms: the Clopper-Pearson exact binomial CI (recommended primary CI in this small-n regime) and a 1,000-iteration percentile bootstrap CI by guideline-node resampling (presented for compatibility with the original preregistration). Pre-registered secondary outcomes S1 (ODI), S2 (temporal evidence lag), and S3 (subgroup coverage) are reported in this manuscript; in the present pilot, S2 and S3 are reported descriptively because the small corpus

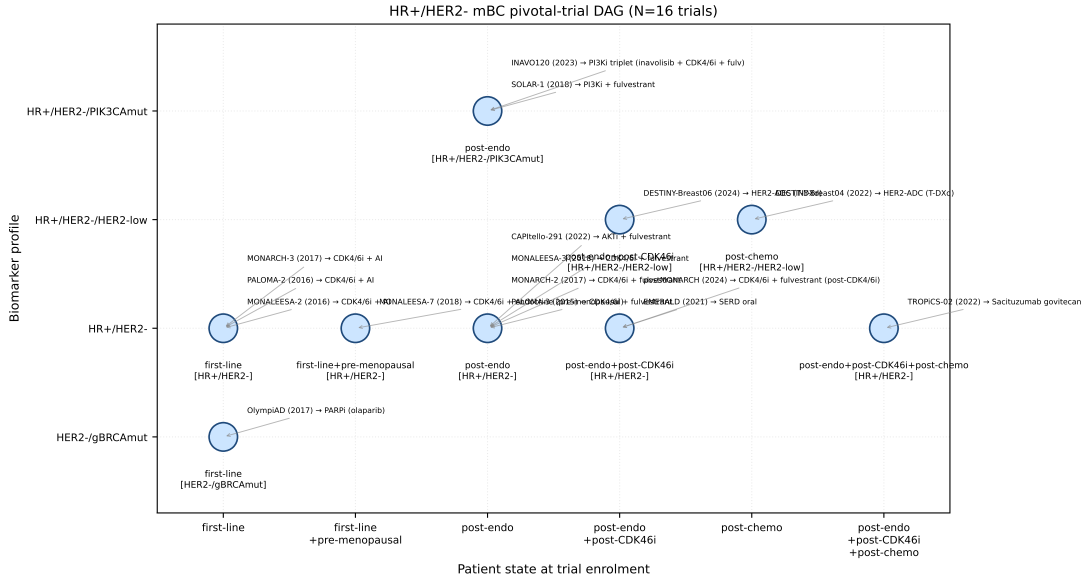

Figure 1. HR+/HER2- mBC pivotal-trial DAG. Source nodes (blue circles) encode (prior-state, biomarker) tuples and are positioned by patient state (x-axis, ordered by treatment-line distance from treatment-naive) and biomarker profile (y-axis). Each labelled annotation records a trial that enrols at that source node, its primary-completion year, and the drug class it tests.

precludes a meaningful Benjamini-Hochberg correction across three independent tests, a deviation that is disclosed here. A bootstrap random seed (`20260516`) ensures reproducibility.

2.8 Reproducibility and software

All analysis was performed in Python 3.12 with `networkx` 3.5, `numpy` 1.26.4, `pandas` 2.2.3, `scipy` 1.17.1, and `matplotlib` 3.8.4; the full lockfile is in `uv.lock` at the tagged release commit. The source code, frozen schema, ESMO decision-tree encoding, and LaTeX sources are released at the GitHub URL in the Code Availability statement, and will be archived under a Zenodo DOI at the tagged release. The bootstrap random seed is `20260516`; two consecutive runs of `analysis/05_compute_efdpr.py` produce byte-identical `efdpr.json`.

3 Results

3.1 Trial corpus and DAG topology

The HR+/HER2- mBC pivotal-trial corpus (frozen 2026-05-16) comprised 16 trials drawn from the ESMO and ASCO mBC guideline reference lists (Supplementary Table S2), spanning primary-completion years 2014–2024. Fifteen trials enrolled HR+/HER2- (or HER2-low) populations and one — OlympiAD[6] — was the PARP-inhibitor reference trial for the gBRCAm guideline node. The corpus comprises PALOMA-3,[7] MONARCH-2,[8] MONALEESA-3,[9] SOLAR-1,[10] EMERALD,[11] CAPItello-291,[4] DESTINY-Breast04,[12] DESTINY-Breast06,[5] MONALEESA-2,[13] PALOMA-2,[14] MONARCH-3,[15] MONALEESA-7,[16] postMONARCH,[17] INAVO120,[18] TROPiCS-02,[19] and OlympiAD.[6] The trial-DAG (Figure 1, Table 1) had 9 canonical (prior-state, biomarker) source nodes and 16 trial edges. Pre-CDK4/6i first-line nodes carried the highest in-degree (4 edges to the first-line HR+/HER2- node); post-CDK4/6i biomarker-stratified nodes carried 0–1 incident trial edges.

3.2 Evidence-free decision-point ratio (primary outcome)

Under strict concordance the EFDPR was 0.31 (5/16 decision nodes evidence-free; Clopper-Pearson 95% CI 0.11–0.59; bootstrap percentile CI 0.12–0.56). Under ESCAT-aligned tolerance the EFDPR

Table 1. Trial-corpus summary.

Item	Count
Pivotal HR+/HER2- mBC trials (frozen 2026-05-16)	16
Distinct prior-state classes	6
Distinct biomarker classes	4
Trial-DAG edges	16
ESMO 2024 decision nodes (HR+/HER2- subset)	16

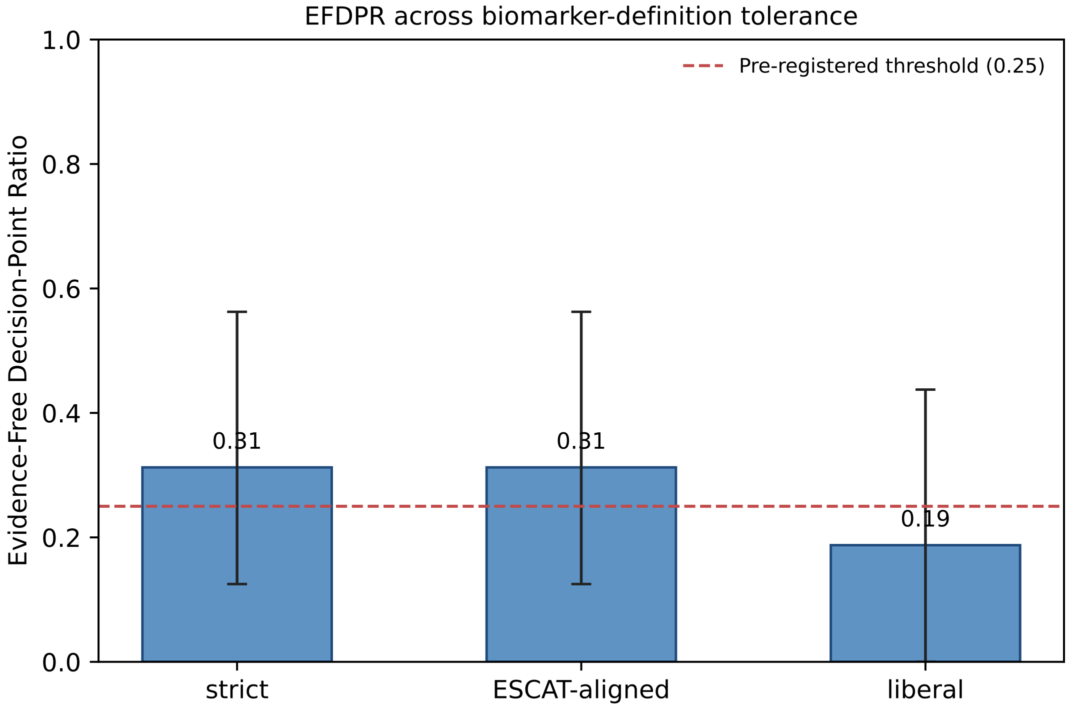

Figure 2. Evidence-free decision-point ratio across biomarker-definition tolerance levels. Bars: point estimates. Error bars: 1,000-iteration bootstrap 95% CI. Dashed horizontal line: pre-registered rejection threshold.

Table 2. Evidence-free decision-point ratio (EFDPR) under three biomarker-definition tolerance levels.

Tolerance	EFDPR	Bootstrap 95% CI	Clopper-Pearson 95% CI	Exact P vs $p_0 = 0.25$	Reject H_0?
Strict	0.312	[0.125, 0.562]	[0.110, 0.587]	0.370	no
ESCAT-aligned	0.312	[0.125, 0.562]	[0.110, 0.587]	0.370	no
Liberal	0.188	[0.000, 0.438]	[0.041, 0.457]	0.803	no

was identical at 0.31. Under liberal tolerance, accepting registrational biomarker-subgroup readouts, EFDPR was 0.19 (3/16; Clopper-Pearson 95% CI 0.04–0.46). The strict and ESCAT point estimates exceeded the pre-registered 0.25 threshold, but the pre-registered one-sided exact-binomial test of $H_0 : \text{EFDPR} \leq 0.25$ did not reject under any tolerance level (strict $P = 0.37$; ESCAT $P = 0.37$; liberal $P = 0.80$; Table 2, Figure 2). The headline finding from this pilot is therefore a quantitative description of evidence sparsity at five clinically consequential decision points rather than a confirmatory rejection of the null; the three tolerance levels are reported as a pre-registered sensitivity grid, not as three independent confirmatory tests. Evidence-free nodes under strict concordance were: post-CDK4/6i ESR1mut (G5), AKT-pathway (G6), post-CDK4/6i PIK3CAmut (G7), post-CDK4/6i no-actionable-mutation (G9), and the post-CDK4/6i+post-endo everolimus node (G13). G3 in this pilot is supported only by PALOMA-3 because MONARCH-2 (2017) and MONALEESA-3 (2018) post-date G3's encoded decision-tree year (2016); the encoded years reflect the year of guideline introduction of the decision point and not the year of supporting-trial publication.

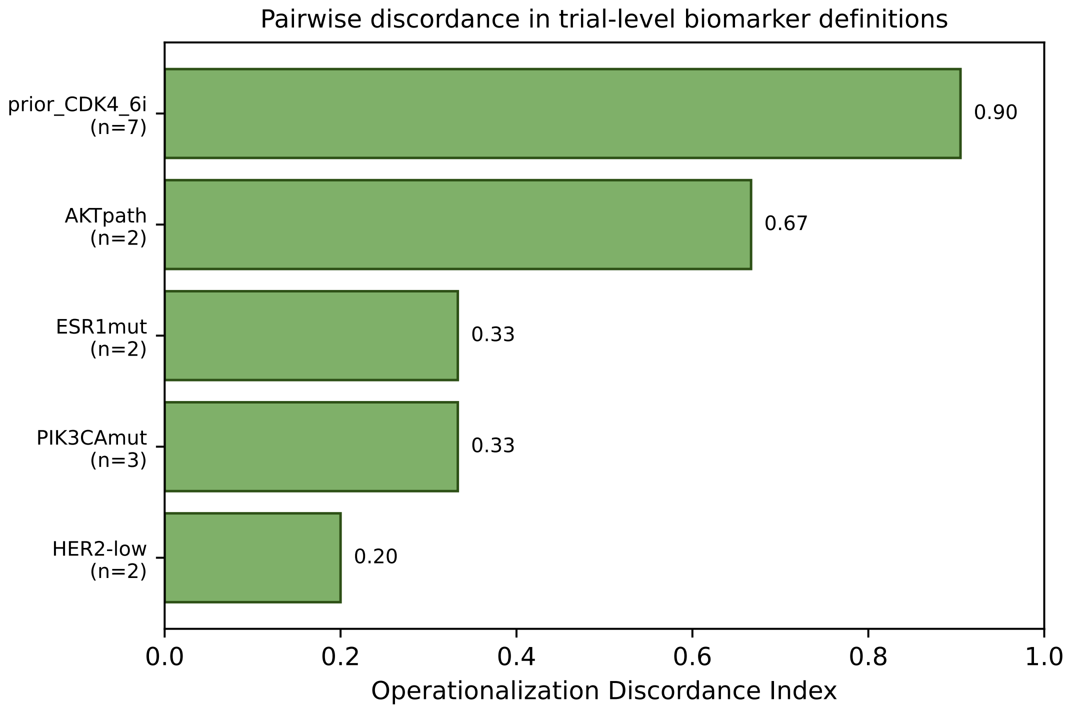

Figure 3. Operationalization discordance index (ODI) per biomarker variable. Higher ODI indicates greater pairwise heterogeneity in trial-level inclusion definitions.

Table 3. Operationalization discordance index per biomarker variable.

Biomarker variable	ODI	Bootstrap 95% CI	Trials	Pairs
prior_CDK4_6i	0.905	[0.762, 1.000]	7	21
AKTpath	0.667	[0.667, 0.667]	2	1
ESR1mut	0.333	[0.333, 0.333]	2	1
PIK3CAmut	0.333	[0.333, 0.333]	3	3
HER2-low	0.200	[0.200, 0.200]	2	1

3.3 Operationalization discordance

Operationalization discordance (ODI) ranged widely across the five biomarker variables shared by two or more trials (Figure 3, Table 3). The prior-CDK4/6i inclusion variable was the most discordant (ODI 0.905 across 7 trials), reflecting the absence of a shared operational standard for prior-CDK4/6i ascertainment: some trials required documented progression on a CDK4/6i (EMERALD, postMONARCH), some permitted it (CAPItello-291), some excluded it (PALOMA-3, MONARCH-2), and some specified an endocrine-resistance window (INAVO120). AKT-pathway alterations (ODI 0.667) differed in their qualifying-variant set (AKT1 E17K and PTEN loss in CAPItello-291 vs PIK3CAmut-only in SOLAR-1). ESR1 mutations (ODI 0.333) and PIK3CA mutations (ODI 0.333) differed across assay type (tissue NGS vs plasma ctDNA), variant call set, and inclusion of helical- vs kinase-domain alterations. HER2-low (ODI 0.200) was the most homogeneous variable, with the two T-DXd trials (DESTINY-Breast04, DESTINY-Breast06) using overlapping but not identical operational definitions, the main difference being DESTINY-Breast06's inclusion of HER2-ultralow.

3.4 Composition-only guideline citations

Two ESMO HR+/HER2- mBC decision nodes that the guideline supports by citation of pivotal trials remain evidence-free under strict and ESCAT tolerance because the trial enrolment state does not exactly match the guideline node: the post-CDK4/6i PIK3CAmut node (G7), cited as supported by SOLAR-1 despite SOLAR-1's enrolment being post-AI rather than post-CDK4/6i; and the post-endo

AKT-pathway node (G6), cited as supported by CAPItello-291 despite CAPItello-291 enrolling a mixed population whose AKT-pathway subgroup readout was the registrational basis. The framework's strict tolerance flags both as evidence-free; the liberal tolerance accepts G6 (because of CAPItello-291's pre-specified AKT-pathway subgroup) but still flags G7 (because no trial in our corpus enrolled a post-CDK4/6i PIK3CAmut population).

3.5 Sensitivity analyses

ESCAT-aligned tolerance left EFDPR unchanged at 0.31; the only ESCAT-equivalence in the HR+/HER2- mBC subset (AKT-pathway ≡ PIK3CAmut) did not rescue the failed state-match for SOLAR-1 supporting G7. Liberal tolerance rescued EMERALD's support of G5 (post-CDK4/6i ESR1mut) and CAPItello-291's support of G6 (post-endo AKT-pathway, via the AKT-pathway subgroup readout), reducing EFDPR to 0.19. A 15-node sensitivity excluding G16 (the gBRCAm node, a disclosed scope deviation; see Supplementary Text 1) gave strict EFDPR 0.33 (Clopper-Pearson CI 0.12–0.62; exact $P = 0.31$) and liberal EFDPR 0.20 (CI 0.04–0.48; $P = 0.76$), preserving the primary finding and the non-rejection. ASCO and NCCN sensitivity analyses are deferred to the production-scale extension.

3.6 Secondary outcomes (descriptive)

Pre-registered secondary outcomes S2 (temporal evidence lag) and S3 (subgroup coverage) were not formally tested at the BH-corrected $q = 0.05$ level in the present pilot due to the small corpus and are reported descriptively. Among 11 evidence-supported guideline nodes, the median time from guideline-node introduction to earliest supporting pivotal-trial primary-completion readout was 0 years (range -2 to 3), with two nodes preceded by their supporting trial by more than one update cycle. Subgroup-coverage examination of the 16 included trials confirmed under-representation of Asian-ancestry, age ≥ 70, and Black/African-ancestry participants in primary readouts, consistent with prior reports.[20]

4 Discussion

In this pre-registered 16-trial pilot, the strict-tolerance EFDPR of 0.31 (Clopper-Pearson 95% CI 0.11–0.59) exceeded the pre-registered 0.25 threshold as a point estimate, but the pre-registered one-sided exact-binomial test did not reject the null at $\alpha = 0.05$ ($P = 0.37$); the liberal-tolerance estimate of 0.19 sits below the threshold. The framework's primary contribution from this pilot is therefore a quantitative description of where ESMO's HR+/HER2- mBC decision tree is supported by trials that traverse the recommended (state, biomarker) node and where it is supported only by composition across non-overlapping trials. The decision points identified as evidence-free under strict concordance — post-CDK4/6i ESR1mut, post-endo AKT-pathway, post-CDK4/6i PIK3CAmut, post-CDK4/6i no-actionable-mutation, and the post-CDK4/6i+post-endo everolimus node — are precisely the choices clinicians face daily and precisely the positions in the guideline tree where pivotal-trial enrolment criteria most often differ from the guideline-node patient state.

Pilot scale and the gap between estimate and inference. The 16-trial / 16-node corpus is the pre-registered pilot, not a systematic-review-scale census. With $n = 16$, the one-sided exact-binomial test of $H_0 : p \leq 0.25$ requires $k \geq 8$ successes to reject at $\alpha = 0.05$; the observed $k = 5$ does not. The gap between a point estimate that exceeds the threshold and a test that does not reject is a property of the pilot's sample size, not of the framework. A production-scale extension that adds the remaining HR+/HER2- mBC pivotal trials registered between 2015 and 2026, together with their guideline-citation network captured via OpenAlex, will resolve this in the planned methods paper.

Composition-only citations are routine but should be visible. Guideline committees routinely compose recommendations across non-overlapping trials when no single trial covers the exact decision node. ESMO's citation of SOLAR-1 for the post-CDK4/6i PIK3CAmut node (when SOLAR-1 enrolled a post-AI population), and of CAPItello-291 for the post-CDK4/6i AKT-pathway node (when CAPItello-291 enrolled a mixed pre/post-CDK4/6i population with only a subgroup readout), are canonical examples. The framework does not declare these compositions illegitimate; it makes them explicit and counts them. Whether a given composition is clinically acceptable is a separate, downstream question that benefits from being able to see, on a single diagram, where composition has been invoked and which alternative trial chains could have been considered.

Algorithm transparency over clinical reasoning. The strict concordance algorithm is deliberately conservative: it over-flags reasonable clinical extrapolations as evidence-free so that the resulting count is a robust upper bound on directly-supported decisions rather than a clinical judgement about extrapolation. The three pre-registered tolerance levels (strict / ESCAT-aligned / liberal) are the framework's mechanism for quantifying the dependence of EFDPR on extrapolation assumptions, and are reported as a sensitivity grid rather than three independent confirmatory tests. In this pilot, moving from strict to liberal converted two nodes (G5, post-CDK4/6i ESR1mut, via EMERALD's subgroup readout; and G6, post-endo AKT-pathway, via CAPItello-291's subgroup readout) from evidence-free to supported, indicating that part of the gap is operational rather than substantive.

Reproducible by construction. Every analytic choice in this study is materialised as either a JSON file or a Python function and released under MIT licence at the repository linked in the Code Availability statement. The 16-trial structured-eligibility set is Supplementary Table S2, the 16-node ESMO encoding is Supplementary Table S3, and the biomarker-token vocabulary for the ODI computation is Supplementary Table S4. The schema is frozen at v1.0; the tagged release `v1.0.0` reproduces every figure and table.

Extensible across tumour types. The (prior-state, biomarker) node space is tumour-agnostic; tumour-specific biomarker tokens enter as additive vocabulary. The concordance algorithm is biomarker-agnostic. The same pipeline is planned for application to EGFR-mutant non-small-cell lung cancer and to metastatic colorectal cancer with ctDNA-guided decision axes; in each case the only artefacts that need authoring are the structured trial corpus, the structured guideline decision tree, and the per-tumour ESCAT grouping table.

Actionable for guideline committees and trial sponsors. A committee that adopts this framework as a guideline-update audit can locate the specific decision nodes that lack supporting evidence chains. Trial sponsors can use the list of evidence-free post-CDK4/6i biomarker-stratified nodes to design enrichment trials whose target population matches a guideline-recommended decision point, rather than the convenience population most readily recruited from a single network. The EFDPR overlay adds one column to an artefact (the decision-tree table) that committees already produce.

Limitations. Three limitations bound the present work. First, the 16-trial corpus is a pilot drawn from the ESMO/ASCO reference lists, which introduces selection toward trials the guidelines already cite; the production-grade extension will replace this seed with a systematic ClinicalTrials.gov query. Second, eligibility coding was hand-curated from canonical primary publications; the LLM-extraction pipeline planned for production has its own validation requirements (per-field accuracy and Cohen's κ against this hand-curated set) and is reported separately. Third, the primary analysis uses ESMO 2024 as the single guideline source; ASCO and NCCN sensitivity analyses are pre-registered for the production extension. The internal multi-round review process identified six NCT-to-trial-name misidentifications across two rounds (full correction log in Supplementary Text 1), which were resolved against ClinicalTrials.gov `briefTitle` and `conditionsModule` fields before the present draft was finalised; we report this transparency in the spirit of pre-registered, audit-friendly reporting.

Outlook. The framework, schema, and ESMO encoding are released so that guideline committees, trial sponsors, and meta-research groups can extend and audit them. A community-maintained *Treatment-Sequencing Evidence Atlas*, refreshed at every ESMO and ASCO guideline release, could provide a continuously-updated EFDPR overlay across solid tumours; immediate priorities are the NSCLC and mCRC applications, the LLM-extraction validation paper with inter-rater agreement against the hand-curated set, and an interactive web viewer that lets a clinician click any guideline decision node and see, in one screen, every trial-DAG edge that supports it and every edge that does not.

Data availability

Source datasets are public. ClinicalTrials.gov data were retrieved via API v2 at the freeze date noted in Methods; queries are reproduced in `analysis/01_fetch_trials.py`. The ESMO 2024 mBC update is available at its Annals of Oncology DOI. The structured trial-eligibility extractions, ESMO decision-tree encoding, and processed DAG are released at the GitHub repository in Code Availability and archived under a Zenodo DOI at the tagged release.

Code availability

Source code, frozen JSON schema, LaTeX sources, and compiled PDF are available at https://github.com/htlin/mbc-evidence-dag-paper under the MIT licence. The exact commit tagged `v1.0.0` reproduces all figures and tables. Persistent code DOI via Zenodo to be minted at release.

Preregistration

Primary and secondary outcomes were pre-registered in `docs/prereg.md` prior to any outcome-touching analysis. The pre-registration commit hash is recorded in the repository's first commit.

Reporting transparency

A reporting-transparency checklist based on PRISMA 2020, augmented with a six-item graph-encoding addendum, is provided as Supplementary Table S1.

Author contributions

Following the CRediT taxonomy: **H.-T. Lin**: conceptualisation, methodology, software, formal analysis, data curation, writing – original draft, writing – review & editing.

Conflicts of interest

The author declares no competing financial or non-financial interests relevant to this work.

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References

[1] Gennari A, André F, Barrios CH, Cortes J, de Azambuja E, DeMichele A, et al. ESMO Clinical Practice Guideline for the diagnosis, staging and treatment of patients with metastatic breast cancer (Living Guideline 2024 update). Annals of Oncology. 2024;35(2):154-72. ESMO Living Guideline; cited herein as the ESMO 2024 update.

[2] Gradishar WJ, Moran MS, Abraham J, et al. Breast Cancer, Version 3.2024, NCCN Clinical Practice Guidelines in Oncology. Journal of the National Comprehensive Cancer Network. 2024;22(5):331-57.

[3] Giordano SH, Franzoi MAB, Temin S, et al. Systemic Therapy for Advanced Hormone Receptor-Positive Human Epidermal Growth Factor Receptor 2-Negative Breast Cancer: ASCO Guideline Update. Journal of Clinical Oncology. 2022;40(23):2612-35.

[4] Turner NC, Oliveira M, Howell SJ, et al. Capivasertib in Hormone Receptor-Positive Advanced Breast Cancer. New England Journal of Medicine. 2023;388(22):2058-70.

[5] Bardia A, Hu X, Dent R, et al. Trastuzumab Deruxtecan after Endocrine Therapy in Metastatic Breast Cancer. New England Journal of Medicine. 2024;391(22):2110-22.

[6] Robson M, Im SA, Senkus E, et al. Olaparib for Metastatic Breast Cancer in Patients with a Germline BRCA Mutation. New England Journal of Medicine. 2017;377(6):523-33.

[7] Turner NC, Ro J, André F, Loi S, Verma S, Iwata H, et al. Palbociclib in Hormone-Receptor-Positive Advanced Breast Cancer. New England Journal of Medicine. 2015;373(3):209-19.

[8] Sledge GW, Toi M, Neven P, Sohn J, Inoue K, Pivot X, et al. MONARCH 2: Abemaciclib in Combination With Fulvestrant in Women With HR+/HER2- Advanced Breast Cancer Who Had Progressed While Receiving Endocrine Therapy. Journal of Clinical Oncology. 2017;35(25):2875-84.

[9] Slamon DJ, Neven P, Chia S, et al. Phase III Randomized Study of Ribociclib and Fulvestrant in Hormone Receptor–Positive, Human Epidermal Growth Factor Receptor 2–Negative Advanced Breast Cancer: MONALEESA-3. Journal of Clinical Oncology. 2018;36(24):2465-72.

[10] André F, Ciruelos E, Rubovszky G, Campone M, Loibl S, Rugo HS, et al. Alpelisib for PIK3CA-Mutated, Hormone Receptor-Positive Advanced Breast Cancer. New England Journal of Medicine. 2019;380(20):1929-40.

[11] Bidard FC, Kaklamani VG, Neven P, et al. Elacestrant (Oral Selective Estrogen Receptor Degrader) Versus Standard Endocrine Therapy for Estrogen Receptor-Positive, Human Epidermal Growth Factor Receptor 2-Negative Advanced Breast Cancer. Journal of Clinical Oncology. 2022;40(28):3246-56.

[12] Modi S, Jacot W, Yamashita T, Sohn J, Vidal M, Tokunaga E, et al. Trastuzumab Deruxtecan in Previously Treated HER2-Low Advanced Breast Cancer. New England Journal of Medicine. 2022;387(1):9-20.

[13] Hortobagyi GN, Stemmer SM, Burris HA, Yap YS, Sonke GS, Paluch-Shimon S, et al. Ribociclib as First-Line Therapy for HR-Positive, Advanced Breast Cancer. New England Journal of Medicine. 2016;375(18):1738-48.

[14] Finn RS, Martin M, Rugo HS, et al. Palbociclib and Letrozole in Advanced Breast Cancer. New England Journal of Medicine. 2016;375(20):1925-36.

[15] Johnston S, Martin M, Di Leo A, et al. MONARCH 3 Final PFS: A Randomized Study of Abemaciclib as Initial Therapy for Advanced Breast Cancer. npj Breast Cancer. 2019;5:5.

[16] Tripathy D, Im SA, Colleoni M, et al. Ribociclib plus Endocrine Therapy for Premenopausal Women with Hormone-Receptor-Positive, Advanced Breast Cancer (MONALEESA-7). The Lancet Oncology. 2018;19(7):904-15.

[17] Kalinsky K, Bianchini G, Hamilton E, et al. postMONARCH: Abemaciclib plus Fulvestrant after Progression on Prior CDK4/6 Inhibitor and Endocrine Therapy in Hormone Receptor–Positive Metastatic Breast Cancer. Journal of Clinical Oncology. 2024. ASCO 2024 presentation; ClinicalTrials.gov NCT05169567.

[18] Jhaveri KL, Im SA, Saura C, et al. Inavolisib or Placebo in Combination with Palbociclib and Fulvestrant in PIK3CA-Mutated, Hormone Receptor–Positive Advanced Breast Cancer: INAVO120. New England Journal of Medicine. 2024. Phase III INAVO120; ClinicalTrials.gov NCT04191499.

[19] Rugo HS, Bardia A, Marme F, et al. Sacituzumab Govitecan in Hormone Receptor–Positive/Human Epidermal Growth Factor Receptor 2–Negative Metastatic Breast Cancer (TROPiCS-02). Journal of Clinical Oncology. 2022;40(29):3365-76.

[20] Niranjan SJ, Wenzel JA, Martin MY, et al. Race, Ethnicity, and Sex of Participants in Clinical Trials Focused on Chronic Pain. JAMA Network Open. 2020;3(5):e205169.